

Transfer learning

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Similarity and normalisation

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sequence representations

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Pooling and spatial information

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Fit and compare for Transfer learning. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Transfer learning. Proposed explanation: “Frozen means the whole model cannot learn”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Fit a new head independently

A separate frozen extractor is $h = 2 \cdot x_1 + x_2$. Training inputs $[1, 0]$, $[1, 2]$, $[2, 1]$, $[3, 1]$ have labels 0, 0, 1, 1. Compare thresholds 2, 4 and 6 using prediction 1 when $h \geq \text{threshold}$; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation $[2, 0]$ with label 1 and $[0, 2]$ with label 0. Compare with the training-majority baseline, whose tie rule chooses 0.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace the reused pipeline

For input $[4, 2]$, compute h using the original extractor and the class at threshold 5. Identify the frozen weights and bias, the fitted quantity, and why h is not a probability.

My reasoning and evidence

6. W2 Fit and compare

For the four target training records, give the number of mistakes at thresholds 3, 5 and 7 and choose the fitted threshold. Then score that head and the declared constant baseline on V1 and V2. Explain which

data fitted the head.

My reasoning and evidence

7. W3 Diagnose a lost distinction

For the first-count-greater target, explain why the original extractor cannot distinguish U [1, 3] from V [3, 1]. Trace the hand-chosen [1, -1] extractor and threshold 1. Does this show that fine tuning has been trained and independently validated?

My reasoning and evidence

8. W4 Fit a new head independently

A separate frozen extractor is $h = 2 \cdot x_1 + x_2$. Training inputs [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1. Compare thresholds 2, 4 and 6 using prediction 1 when $h \geq \text{threshold}$; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0. Compare with the training-majority baseline, whose tie rule chooses 0.

My reasoning and evidence

9. W5 Plan credible evidence

A report omits the source checkpoint and preprocessing, claims faster learning without timing, and repeatedly chooses settings using the final evaluation set. Name the missing records and propose a fair comparison with a model trained from scratch.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled transfer learning evidence sheet to Image-pattern Network Investigation. Show how you can trace a reused feature extractor and a newly fitted output head. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Transfer learning

Transfer a fixed feature extractor while identifying what is actually refitted.

Frozen extractor $h = \text{left} + 2 \times \text{right}$. New-task training inputs $[1, 0] \text{label} 0, [2, 1] \text{label} 0, [1, 3] \text{label} 1$. Class 1 when $h \geq \text{threshold}$; candidates 3 or 5.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Transfer learning

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Extract features

Training values are 1, 4, 7.



2. Fit new head

Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5.



3. Check held-out input

Query [2, 2] has $h = 6$, so the head predicts class 1; its reference is not supplied.

CHANGE ONE THING

Query changes to [2, 1].

Explain the changed result using the matching lesson.

Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.